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Contextual transformer for brain stroke segmentation in MRI

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ABSTRACT

Accurate segmentation of brain stroke using magnetic resonance imaging (MRI) is associated with difficulties due to the complicated anatomy of the brain and the different properties of the stroke. A well-known challenge associated with this problem is the variability in the size and shape of the stroke, which makes it difficult to employ standard segmentation strategies. This study introduces a contextual transformer framework, which synergies spatial feature extraction with global anatomical processing ability, further enhanced by advanced feature fusion and segmentation synthesis techniques. The main idea is based on focus in and generalize out (FIGO), which uses a special pipeline that focus in stroke features with generalization out external anatomical structures. Experimental studies using the ATLAS v2.0 dataset demonstrate a remarkable improvement in segmentation accuracy using FIGO. Ablation studies confirm the significant impact of the proposed framework in comparison with conventional methods.

1. Introduction

Stroke is one of the most challenging public health problems around the world, as it consistently ranks at the top as the leading cause of mortality and long-term disability in diverse populations [1,2]. The incidence of stroke-related deaths is likely to increase significantly, increasing from 6.6 million cases in 2020 to an estimated 9.7 million in 2050 [3]. The growing number of stroke cases underscores the need for innovative technologies, both diagnostic and therapeutic, that can efficiently deal with this problem, targeting its determinants, symptoms, and effects in terms of disability and death [4,5]. Brain stroke is commonly attributed to two primary causes: ischemic stroke, characterized by blockage of blood vessels, and hemorrhagic stroke, resulting from rupture of vessels leading to bleeding into surrounding tissues (Fig. 1). In neurodiagnostics, the contributions of magnetic resonance imaging (MRI) are irrefutable due to its ability to provide high spatial resolution, allowing brain structures to be precisely illustrated [6]. Accurate identification and characterization of brain regions affected by stroke is a crucial step towards appropriate treatment and prognosis. The spatial resolution of MRI is commonly 1–2 mm for most sequences, which is often enough for most clinical cases [7].

The past decade has been marked by significant milestones in medical image analysis, largely due to the integration of sophisticated computational technologies [8]. Deep learning methods, particularly convolutional neural networks, have guided a new era of automatic medical imaging, revolutionizing conventional segmentation challenges [9,10]. Models such as SwinUNETR provide the visualization of this transition. They are fast, accurate, and scalable solutions that reduce the variability engendered by

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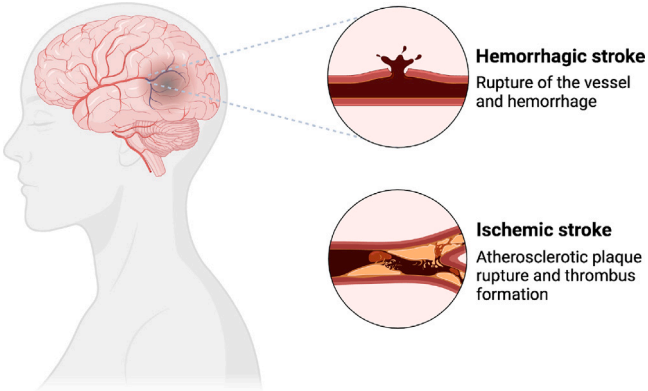

Fig. 1. Different types of brain strokes (ischemic and hemorrhagic).

human interpretation [11,12]. SwinUNETR, an innovative transformer-based model, has achieved the highest level of performance in various medical image segmentation tasks, including the BTMultiorganan Segmentation Challenge and the Medical Segmentation Decathlon (MSD) [13]. For example, such technological advances have proven their importance, even in resource-constrained settings where radiological expertise is limited [14]. Identifying and segmenting the stroke from MRI is a critical aspect in selecting the right clinical interventions and formulating an accurate prediction about the patient's recovery [15]. Fig. 2 shows various types of stroke in different sizes and locations, making it difficult to accurately identify and delineate these pathological phenomena. However, despite these technological advancements, current models still have challenges in accurately detecting stroke lesions of varying sizes within a unified diagnostic framework [16].

Another problem that has been well-known for a long time is the data imbalance, caused by small lesions being underrepresented compared to big ones [17]. Although transformers have performed very well at capturing global contextual information, they usually fail to perform localized, meticulous tasks. Transformers use the self-attention mechanism that focuses on long-range dependencies and global features, with a tendency to overlook fine-grained details [18]. Common deep learning models perform well at recognizing either small, medium, or large stroke lesions, but they can hardly detect all sizes concurrently [19]. This highlights a crucial gap: the need for models that have the flexibility to detect stroke lesions of various sizes simultaneously and with high accuracy.

This paper aims to fill the gap in current research by proposing a novel deep learning framework developed to address the challenge of segmenting ischemic stroke in MRI. Ischemic strokes occur more frequently, which makes correct and timely identification crucial. The proposed model integrates the SwinUNETR architecture [20] with the U-Net framework [21] in a way to enable focus in and generalize out (FIGO) framework. This transformer architecture improves computational performance and increases the feature processing capacity. Its ability to shift the windowing scheme makes it more feasible for medical image analysis. TransUNet employs a sequential approach to integrate CNN and Transformer features. It does not include a specialized mechanism for optimizing the fusion of local and global information. Also, its focus is on 2D segmentation rather than 3D. FIGO introduces an innovative feature fusion strategy and connects the strengths of U-Net's detailed spatial features with SwinUNETR's contextual insights. It allows for a more balanced representation of both local and global features. This enables better handling of the varied and complex morphologies of brain stroke. The contributions of this study are:

- Develop a deep learning framework that improves the precision of brain stroke segmentation in MRI,
- Solve the data imbalance problem by creating a comprehensive pre-processing pipeline that also improves the model's ability on different stroke sizes and shapes,
- Validate the model effectiveness on the challenging ATLAS v2.0 benchmark dataset in comparison with conventional methods.

This paper is organized as follows: related work investigates the current state of segmentation methodologies and deep learning for medical images. The methodology describes the preprocessing pipeline along the FIGO framework. Training protocols and the experiment results show the model implementation details and results, followed by a discussion and conclusion with future directions [22,23].

2. Related work

Several approaches have been explored, and the spectrum ranges from algorithmic developments to advanced deep learning models. Gonzalez et al. [24] have developed the Fuzzy Information Seeded Region Growing (FISRG) algorithm that incorporates fuzzy logic along with other regional growing methods to improve the segmentation performance. Liu et al. [25] proposed HCSNet, a hybrid contextual semantic network that combines both spatial and channel contextual features, based on the encoder–decoder structure for segmenting small lesions. Further, Liu et al. [26] designed a multi-task learning strategy called SQMLP-net, where segmentation and classification tasks are combined to improve feature sharing and optimization. Several shapes and forms of U-Net

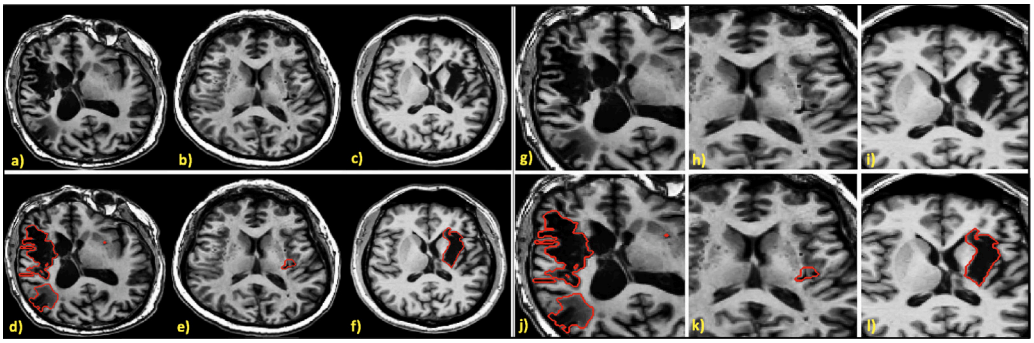

Fig. 2. The figure shows the challenges associated with segmentation in diverse stroke cases: (Left) The original MRI slices (a), (b), and (c) show strokes of varying sizes and locations. The ground truth mask overlays presented in (d), (e), and (f) demonstrate the intricacy of precisely segmenting these strokes. (Right) While the (g), (h), and (i) provide cropped views of anatomical complexities, the (j), (k), and (l) provide detailed ground truth masks. It highlights areas that are challenging to delineate and the inherent segmentation difficulties.

models were tested by Deb et al. [27], specifically about 2D and 3D MRI segmentation results. Chalcroft et al. [28] went beyond developing U-Net designs by including large-kernel attention mechanisms to enhance the interaction of the networks within large fields of view.

To address the issue of intensity variation, Huo et al. [29] introduced the Adaptive Region Harmonization Network (ARHNet), where regions are harmonized adaptively. Mohapatra et al. [30] conducted a meta-analysis on transfer learning in brain lesion segmentation and concluded that both mixed data approaches and intermediate task training enhance performance. Zafari-Ghadim et al. [31] present a recent review on transformer-based architectures for stroke segmentation. Rachmadi et al. [32] proposed the 3D residual U-Net and the instance-wise Center-of-Instance (ICI) loss function to solve the instance imbalance of biomedical segmentation. Huo et al. [33] presented an nnU-Net-based segmentation framework that utilizes ensemble learning and post-processing strategies for enhanced performance. Verma et al. [34] developed an automated segmentation model based on 3D-UNet for identifying chronic brain strokes. Wu et al. [23] developed MLiRA-Net, a stroke lesion segmentation architecture from the multi-scale transformer-convolutional hybrid structure with long-range interactive and regional attention. Wang et al. [22] introduced a stroke lesion segmentation model from convolutional fusion and an improved attention mechanism.

Recent studies emphasized hybrid CNN–Transformer networks to promote stroke lesion segmentation based on local spatial details and global context. Kuang et al. [35] proposed a parallel hybrid encoder as a composition of CNN and transformer, along with a circular feature interaction mechanism for information exchange. This model invokes bilateral difference learning for hemispheric asymmetry capture in stroke lesions in CT modalities. Ahmed et al. [36] introduced deformable attention blocks in CNNs and lesion-insertion data augmentation, and attained good dice performance in the case of small or irregular lesions in MRI. The experiments reflect the merit of attention-based architectures, which is consistent with our suggested FIGO architecture that seamlessly integrates swin transformer and spatial CNN building blocks in sequence for 3D MRI lesion delineation. Recent work addresses complementary challenges in stroke-focused AI as well as segmentation-based work. Soladoye et al. [37] reviewed ML-based stroke forecasting pipelines with a focus on generalization, dataset bias issues, and concerns. Prick et al. [38] defended the interpretability of systems in clinical decision-making in support of explainable and robust models.

Recently, there have been notable developments in brain stroke segmentation using MRI scans. The ATLAS v2.0 dataset [39] is a key tool for evaluating and benchmarking new methodologies. Table 1 summarizes previous studies, including the aim, dataset insights, preprocessing methods, structure, architecture descriptions, and loss functions. Table 1 also includes the year each segmentation model was proposed, making it clear how it has evolved across architectures. This compilation is not only an exhibit of the variety and complexity of modern approaches but also a contributor to the deepening of knowledge regarding this subject. The studies are stacked together to form the foundation of the model's methodology. Following the technical explanation in Table 1, there is a need for further elaboration on the major findings and methodological innovations in these studies. It provokes an in-depth study, and Table 2 is supplemented with more detailed information on the innovations and deficiencies of these recent studies. This comprehensive summary serves as the basis for the proposed framework.

This architecture innovation adheres to a prevailing trend in the literature to blend multi-scale attention, global tokenization, and axial context. Li et al. [40] developed SFMANet, which integrates spatial-frequency gating and multi-scale axial attention to enhance lesion boundary sharpening and suppress over-smoothing. Wu et al. [41] developed TransRender, a transformer network that keeps decoding attention concentrated in uncertainty regions of edges, iteratively refining ambiguous edges of lesions. Both methods place special emphasis on contour localization accuracy, which is well in line with our FIGO architecture's focus on combining spatial as well as global anatomical information. Our realization integrates these contexts in an embedded swin transformer, which is governed through fused encoder–decoder interactions.

However, some limitations are still inherent in the study of stroke lesion segmentation. Several current models that Deb et al. [27] assessed for improvement and Rachmadi et al. [32] also stressed the need to improve on the generalization of the model to different sizes and types of lesions. These studies also concentrate on enhancing the handling of imbalances in biomedical images.

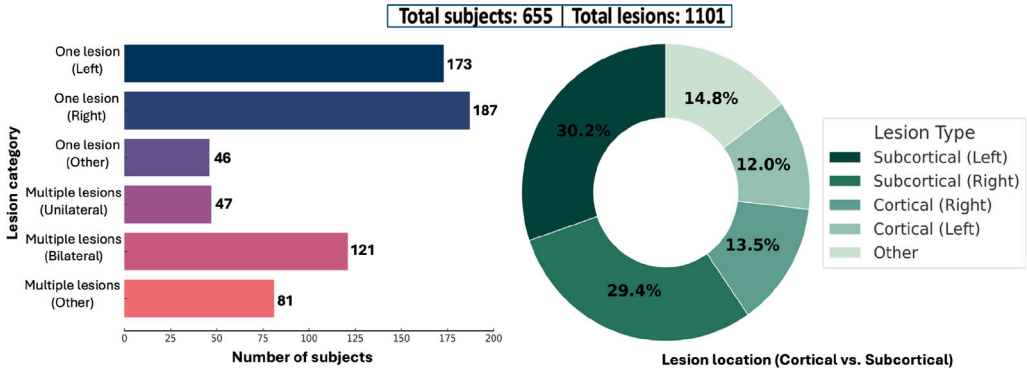

Fig. 3. Stroke distribution in the ATLAS v2.0 training dataset. The bar plot shows the number of subjects with either single or multiple lesions, subdivided into specific hemisphere locations. The pie chart shows the percentage of lesions identified in different regions.

Furthermore, Gonzalez et al. [24] and Huo et al. [29] demonstrate that lesion segmentation becomes challenging when lesion intensities are similar to those of surrounding areas. Chalcroft et al. [28] and Liu et al. [26] have applied transformer models that can potentially capture global context to achieve multi-task learning optimization. Thus, refinements are still required on kernel size optimization, patch embeddings, and task weight balancing. As observed by Mohapatra et al. [30] in transfer learning techniques, similar analyses are also necessary for scalability in 3D performance measures. In addition, Huo et al. [33] and Verma et al. [34] indicate that automated models require significant attention to biases in training data and independent evaluations of models. The models stated previously fail to incorporate both spatial and global contextual features effectively. In the proposed FIGO framework, the feature fusion and segmentation synthesis process is incorporated in a way that enhances segmentation precision and stability while addressing the flaws identified in current approaches.

Previous research shows self-supervised pre-training can overcome annotation inadequacy, promoting model robustness and generalizability. Soh and Rajapakse [42] introduced a hybrid U-Net–Transformer network, which is pre-trained under noise-anchored self-supervision for effective segmentation under limited labeling. Future enhancements may include a self-supervised pretext task or domain-adaptive fine-tuning for further use in resource-constrained settings. Such observations motivate the choice of architecture in the proposed FIGO framework, which attempts to harmonize spatial encoding and global contextual perception through transformer-guided fusion.

3. Methodology

The FIGO framework is designed to enhance the accuracy of lesion segmentation in MRI. This section explains the dataset description, pre-processing pipeline, and proposed network architecture.

3.1. Dataset

The ATLAS v2.0, a well-curated dataset of 3D T1-weighted MRI scans [39]. The dataset is the extended version of ATLAS v1.2, comprising 1271 MRI subjects with manually labeled strokes. The dataset is divided into three subsets: (1) Training set: comprising 655 MRIs with ground truth values; (2) Testing set: which includes 300 MRIs without labels; and (3) Generalization set: contains 316 MRIs with data not previously exposed during the training phase, testing the model's ability to generalize. The dataset is diverse and spans forty-four cohorts and eleven countries, providing a robust representation of the various types of lesions, their locations, and sizes. The diversity will result in creating a model that can be broadly applied to various clinical conditions. The dataset incorporates a range of stroke sizes, locations, and appearances. It is compiled to cover a wide spectrum of stroke features, including single and multiple strokes in different cerebral hemispheres (Fig. 3). Another challenge is brain anatomy's complexity and the multispectral features of strokes. The complexity is a step higher in 3D images, where the edges of the lesions from the surrounding brain tissue become particularly important in differentiation. This causes difficulty in identifying lesions accurately and segmenting them using simple methods. Also, manual lesion segmentation is a subjective process that may lead to a bias in the training dataset. This study employs the ATLAS v2.0 training dataset, which includes 655 MRI subjects. Due to the unavailability of ground truth values in the testing subset, the training is randomly split into two sets (80%, 20%) for training and testing, respectively. Lesion sizes in the ATLAS v2.0 dataset vary significantly. We define small lesions as <5000 mm^3, medium lesions as 5000–20,000 mm^3, and large lesions as >20,000 mm^3, based on clinical thresholds. The dataset includes 354 small (54.0%), 111 medium (16.9%), and 190 large (29.0%) lesions out of 655 subjects. Our FIGO framework segments lesions of any size through a unified architecture. Qualitative results (Figs. 7 and 8) show that FIGO accurately segments small, medium, and large lesions.

Table 1
Overview of brain stroke segmentation methods utilizing ATLAS v2.0 dataset. The first three studies were newly added in this revision to expand the comparative context with recent methods.

Purpose	Dataset	Preprocessing	Methods	Structure	Loss function	Year	Ref.
Improve stroke lesion segmentation using cross-scale attention to enhance small-lesion detection	ATLAS v2.0	MNI-152 registration, patch-based input (128^3)	Multi-stage cross-scale attention (MSCSA) integrated into U-Net skip paths	U-Net with MSCSA module and Intra-FFN blocks	Dice + cross-entropy	2025	[43]
Improve stroke lesion segmentation using deformable attention and lesion-insertion augmentation in a DETR-style transformer model	ATLAS v2.0	MNI-152 registration, isotropic resampling, and data augmentation	Transformer-based DETR architecture with deformable attention and multi-task heads	CNN backbone with transformer decoder for segmentation, detection, and refinement	Compound loss: Dice, CE, Focal, L1, GIoU, and point-based loss	2024	[36]
Improve small-lesion segmentation using volume- and boundary-aware label reweighting	ATLAS v2.0	MNI-152 registration, z-score normalization, patch extraction (128^3)	Multi-size labeling (MSL) and distance-based labeling (DBL) with ensemble and postprocessing	Standard 3D U-Net with label-aware loss handling; no architectural change	Dice + cross-entropy	2024	[44]
Benchmark various deep supervised models for stroke lesion segmentation	ATLAS v2.0	Z-score normalization, slice handling for 2D models	Evaluation of deep supervised U-Net style models on 2D and 3D MRI	Multiple U-Net variants (traditional, residual, and attention-based)	Unspecified	2024	[27]
Improve the segmentation of biomedical images by addressing instances of imbalance	ATLAS v2.0	Intensity normalization, data augmentation (flipping, rotation, zooming, shifting)	Combined instance-wise and center-of-instance loss with comparisons to dice loss and blob loss	3D Residual U-Net	ICI loss combining global dice, instance-wise, and center-of-instance losses	2024	[32]
Improve the accuracy of stroke lesion segmentation in MRI using a novel FISRG algorithm	ATLAS v2.0	Gaussian denoising	Fuzzy Information Seeded Region Growing (FISRG) with k-means clustering for seed selection and mathematical morphology for post-processing	Combines fuzzy logic with SRG techniques	Unspecified	2023	[24]
Segment small-size stroke lesions from MRIs using HCSNet	ATLAS v2.0	2D image slicing, matrix complement, clipping	Hybrid Contextual Semantic Network (HCSNet) with an encoder–decoder architecture and HCSM	U-shaped architecture with HCSM	Dice + focal loss	2023	[25]
Joint learning for stroke lesion segmentation and TICI grading using SQMLP-net	ATLAS v2.0	Intensity correction, MNI-152 template registration, and normalization	SQMLP-net with a segmentation branch and a classification branch	Hybrid multi-task network with shared encoder	Joint loss function combining segmentation and classification losses	2023	[26]
Enhance brain lesion segmentation using large-kernel attention within a U-Net architecture	ATLAS v2.0, ISLES, and BraTS	ATLAS: skull-stripping, bias-correction, re-slicing; ISLES: re-slicing, augmentation; BraTS: re-slicing, skull-stripping, cropping	Convolutional transformer block variant in U-Net architecture with large-kernel attention and post-processing	U-Net with transformer blocks	Dice and cross-entropy loss	2023	[28]
Improve stroke lesion segmentation accuracy through adaptive image harmonization	ATLAS v2.0	Normalization, data augmentation, and perturbation techniques	Adaptive region harmonization (ARH) for foreground and background alignment	ARHNet architecture	Reconstruction, boundary-aware total variation, and adversarial loss	2023	[29]

(*continued on next page*)

Table 1 (*continued*).

Meta-analysis of transfer learning for brain lesion segmentation	ATLAS v2.0, In-house dataset	Resampling, skull stripping, image slicing, and data augmentation	Mixed data approach and intermediate task training (ImTT) using transfer learning	Various 2D deep learning architectures	Unspecified	2023	[30]
Improve the generalization ability of stroke lesion segmentation using deep learning	ATLAS v2.0	Registered to the MNI-152 template, normalization	nnU-Net based model with various training schemes, ensemble techniques, and post-processing	nnU-Net, an encoder–decoder architecture	Dice + cross-entropy	2022	[33]
Improve the chronic stroke lesion segmentation	ATLAS v2.0	Intensity normalization, MNI-152 template registration, HD-BET, and data augmentation	Deep Neural Network (DNN) using 3D-UNet with 5-fold cross-validation	3D-UNet for volumetric segmentation	Unspecified	2022	[34]

3.2. Data pre-processing pipeline

The model's accuracy is highly dependent on a data pre-processing infrastructure that is systematically designed. The pipeline applies a series of operations to the raw MRI to make it suitable for reliable feature extraction. This pipeline unifies the N4Bias field correction and adaptive resampling methods to ensure the input data is high-reliability and consistent.

3.2.1. Resampling

It began by loading an MRI dataset with the help of the `load_nifti` function in NIfTI format to preserve data integrity and consistency. One of the core steps here is substituting zeros for NaN (not a number) values to ensure smooth model training. This is accomplished through the `replace_nans_with_zero` function, which examines each image for NaN values and replaces them with zeros. This method guarantees data consistency and prevents computational errors that could disrupt subsequent analyses. Then, the subjects are resampled to standardized voxel dimensions, a vital step to ensure compatibility with different scanning protocols and equipment. This standardization is done by using custom-made functions, which are used to guarantee unambiguity across the dataset. Two interpolation techniques are employed for resampling: 'nearest' interpolation for binary masks that helps preserve lesion boundaries and 'linear' interpolation for image data that promotes continuity of intensity values. This process aims to preserve the accuracy of the spatial model of the brain, which means that for all the subjects, spatial dimensions are uniform.

3.2.2. Bias correction

We employ the N4Bias Field Correction algorithm [45], a technique focusing on intensity variation correction across images. Magnetically inhomogeneous intensities of magnetic fields can also be seen in MRI due to different scanner calibrations and magnetic fields. These heterogeneities may not provide clear details of the images and make it difficult to differentiate pathological features. The N4Bias field correction algorithm uses a nonlinear filtering method. The function `bias_field_correction` utilizes SimpleITK's `N4BiasFieldCorrectionImageFilter`, a non-linear image filtering process. This process contributes to reducing the impact of external variability in data. The corrected pictures display improved clarity and uniformity.

3.2.3. Data standardization and augmentation

This last step of the pre-processing pipeline highlights the necessity of image normalization and a data augmentation strategy. The normalization of MRI to the range of [0, 1] is accomplished by scaling the intensity values. The scanning is performed on an image-to-image basis to ensure that there is a single, unified dataset. This scaling not only makes uniform input distribution simpler, but it also improves the model's ability to detect subtle differences in tissue characteristics. Normalization is vital to make the model work with data having the same statistical properties, which are heavily based on gradient-based optimization techniques. This step also includes error handling to ensure robustness and reliability, particularly when image retrieval or processing fails. Following normalization, image resizing is conducted to a consistent size of $160 \times 160 \times 160$. This standardization of all images implies that the images have a uniform size. A standardized input size facilitates the model's learning process because of the uniformity of the input size. Moreover, the data augmentation technique formulated by the MONAI framework[1] is employed to synthetically enlarge the dataset by simulating natural variations. Methods like random flipping, rotation [46], and affine transformations can simulate the disorder observed in real-world anatomical structures. Each transformation is carefully designed to allow the model to generalize from training data to previously unseen clinical images.

[1] https://monai.io/.

Table 2
Contributions, highlights, and identified research gaps of related work are listed in Table 1. The first three studies were newly added in this revision to expand the comparative context with recent methods.

Contribution	Highlights	Gap	Ref.
Proposed MSCSA module to improve the mapping between the brain structural features and lesions of varying sizes using cross-scale attention and Intra-FFN for stroke lesion segmentation.	Achieved consistent gains on small-lesion segmentation while maintaining stable results on the full dataset.	Performance on medium and large lesions remained comparable; further improvement in global representation is needed.	[43]
Proposed a 2D transformer model with deformable attention and lesion-insertion augmentation for stroke lesion segmentation.	Achieved good results across lesion sizes, with top scores in medium and large groups.	Operates in 2D; lacks volumetric contextual capability.	[36]
Introduced multi-size and boundary-aware label restructuring (MSL + DBL) to improve small-lesion segmentation without modifying the architecture.	Demonstrated improvement on the small lesions subset and matched baseline ensembles on the full dataset.	Method relies on manually defined lesion size and boundary thresholds; lacks spatial modeling.	[44]
Conducted a thorough comparison of the different U-Net architecture designs for both 2D and 3D MRI of stroke lesion segmentation.	Pointed out that the U-Net architecture can tackle large datasets and be used in both 2D and 3D modalities.	It indicates the need for a deeper investigation of transformer models in 3D segmentation and the use of the model on large datasets.	[27]
Introduced instance-wise and center-of-instance (ICI) loss to deal with instance imbalance in biomedical image segmentation using a 3D residual U-Net architecture.	The demonstrated ICI loss certainly improved performance, especially in segmenting small instances and addressing imbalances between instances.	More validation of ICI loss's effectiveness is needed for a larger number of segmentation tasks and datasets.	[32]
Developed the FISRG algorithm, which improves stroke lesion segmentation using fuzzy logic with seeded region growing techniques.	Demonstrated high accuracy in processing various appearances of lesions on MRI scans; offers a promising basis for stroke diagnosis.	The algorithm needs to be improved in terms of its capability to discriminate lesions from neighboring areas that have the same intensity and adaptability to the topological changes.	[24]
Proposed the HCSNet, combining spatial and channel contextual features and the design of the encoder–decoder architecture, for a precise segmentation of the small-sized stroke lesions.	The network had demonstrated superior performance in segmenting small lesions, benefiting from the fused contextual features capability.	HCSNet needs to be evaluated in more detail concerning its generalization ability for different lesion sizes and types.	[25]
Introduced the novel method of simultaneous segmentation and grading while leveraging quantum mechanics simulations for learning.	Demonstrated well-performed feature sharing and task optimization, which enabled its multi-task learning performance.	It is necessary to find the right balance between the trade-offs of multi-task learning weights more efficiently.	[26]
Proposed a U-Net model with large kernel attention methods to take advantage of the strengths of CNN and transformer to perform 3D brain lesion segmentation.	The model showed competitive performance with the use of an efficient number of parameters and a special focus on long-range spatial dependencies.	There is a need for research on different kernel sizes as well as embeddings of the patches.	[28]
Introduced the novel ARHNet model that targets the issue of image intensity disparity by integrating intensity perturbation and region-adaptive harmonization.	Demonstrated the superiority of the proposed method in image harmonization to ensure the strength of augmented images.	Differentiating small lesions remains a challenge; understanding the features of different lesions is still a problem.	[29]
Implemented transfer learning and mixed data techniques across 2D deep learning models to improve stroke lesion segmentation accuracy.	The application of transfer learning and ensemble methods increased the accuracy, especially by using the novel agreement window technique.	Further investigation is needed on scalability to 3D models and more detailed performance metrics for specific models.	[30]
Developed a segmentation framework with nnU-Net, including ensemble learning and post-processing techniques to improve the model's generalization.	Demonstrated top performance in the MICCAI ATLAS Challenge, which evaluated the efficacy against various lesion sizes and shapes.	The problem remains for tiny or infrequent spots; future improvements are needed to optimize segmentation accuracy.	[33]
Implemented an automated segmentation model based on the 3D-UNet architecture for chronic stroke lesion detection.	Obtained a proficient level of segmentation accuracy, creating a base for automated evaluation of chronic stroke lesions.	There is a need for research in the field of training dataset bias and for independent evaluation to confirm the model's generalizability capability.	[34]

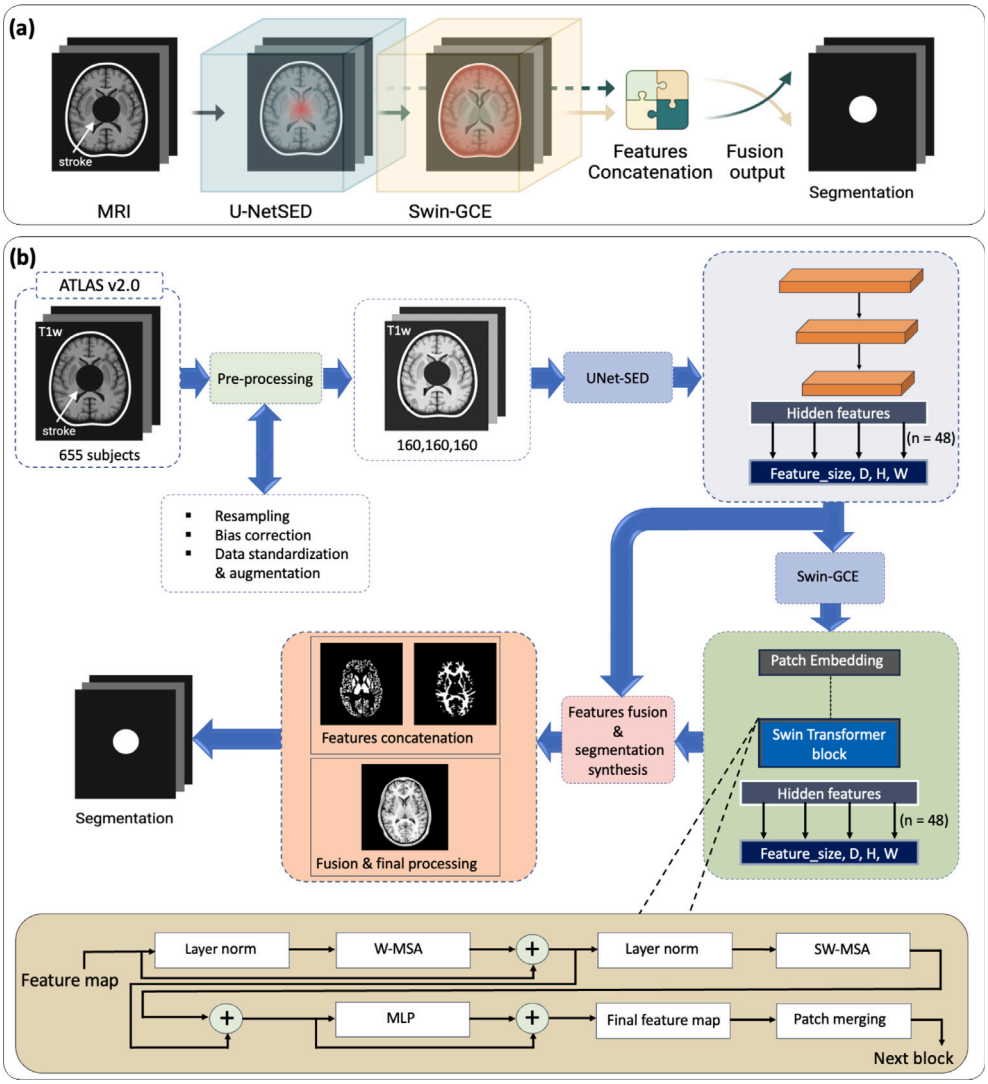

Fig. 4. Detailed overview of the FIGO framework. (a) Semantic visualization emphasizes the synergistic combination of the spatial and contextual information processing components. (b) The detailed workflow diagram shows the model's pivotal components in depth, starting from preprocessing. It also illustrates the flow from U-NetSED for the initial feature extraction, followed by global feature enhancement using Swin-GCE. In this way, the feature fusion, followed by the segmentation synthesis, helps balance the consideration of both local and global features. The process ends with the fusion and final processing, which integrates the outcomes of each segmentation stage and gives us the segmentation result.

3.3. FIGO framework

The proposed FIGO framework (Fig. 4) sequentially integrates the functionality of U-Net with the adaptability of SwinUNETR to effectively manage the challenges of MRI. This integrated procedure combines the local features extracted by U-Net with the global contextual information provided by SwinUNETR. This architecture is designed to inspect brain structures and provide detailed segmentation of brain stroke (Fig. 5).

FIGO framework, unlike the classical methods that apply these models separately, integrates these models into a single comprehensive scheme. The Swin-GCE includes an attention mechanism, making it suited for the high-dimensional data typical of MRI. The neural network architecture becomes more sophisticated by adding an extra depth of layers and employing dropout and batch normalization methods at selected points. It follows a structured pipeline of processing that not only preserves but also improves each model's strengths. These adaptations ensure that the transformer's power in capturing extensive contextual information complements U-Net's precision in local detail extraction.

3.3.1. U-Net spatial encoder–decoder (U-NetSED)

The U-Net structure is symmetric, with the encoder path that holds context and the decoder path that gives high-precision localization. The encoder accomplishes this by using convolution layers to sequentially downsample the input image, extracting features while retaining relevant contextual information at multiple scales. The deep convolutional layers in the architecture have variable kernel sizes, which are intended to optimize the network to meet the different shapes and sizes of stroke lesions.

$$f_i^n(x) = \text{PReLU}\left(BN\left(Conv_n\left(f_i^{n-1}(x)\right)\right) + x\right). \tag{1}$$

At the ith layer, the convolution operation is denoted as $\text{Conv}_i\ (x)$, ranging from 1 to n. Following that, batch normalization is used to normalize each convolution's output. Finally, PReLU is applied to each normalized output after each convolution. To deal with variability in stroke lesions, the U-Net part has been modified by increasing the channel depth and adding some targeted residual connections to optimize its capability. These are designed to enhance the detection and characterization of stroke lesions. Every layer incorporates dropout and batch normalization, which helps the learning process and prevents overfitting. The increased number of channels at each convolutional block allows the architecture to detect a wider range of features, making it sensitive to different lesion characteristics. Residual connections are added to provide a stable gradient flow during the training. This helps to stabilize the training process and allows more complex patterns to be learned without the gradient vanishing. The output feature maps of U-Net become the input for the SwinUNETR, bringing the local spatial features of U-NetSED into the contextual knowledge of Swin-GCE.

3.3.2. SwinUNETR global context encoder (Swin-GCE)

The Swin-GCE component is designed to capture global contextual information. The SwinUNETR architecture takes advantage of the Swin Transformer architecture and complements it with shifted windowing schemes for self-attention. This adaptation helps to keep computational costs down and ensures that dependencies can be captured over long ranges. Unlike traditional transformers that address self-attention through the entire input space, SwinUNETR focuses on self-attention within the local windows. A technological component is to shift the window from layer to layer, ensuring comprehensive coverage and integration of global context throughout the imaging volume. The SwinUNETR's operation in the framework is expressed as follows:

$$\mathcal{F}_{\text{Swin}}(x) = x + \text{MLP}(LN(x + SW - MSA(LN(x)))). \tag{2}$$

The input x denotes the output of the feature map from U-NetSED, which is subsequently transferred to SwinUNETR. The shifted window multi-head self-attention ($SW - MSA$) output is added to the input of the original x, forming a residual connection. The sum is then normalized using layer normalization (LN). Then the MLP block receives the normalized feature set and provides the model with better presentable features. To solve the vanishing gradient problem, a residual connection is added to the MLP output, where the original input x is taken as the addition. SwinUNETR integration following U-NetSED feature map extraction is essential, as U-Net's detailed local features would enable the SwinUNETR to focus on the global context for accurate segmentation. The high processing capacities of SwinUNETR, in conjunction with the refined details of the enhanced feature maps from U-Net, improve the model's performance.

3.3.3. Features fusion and segmentation synthesis

The final part of the architecture is the feature fusion mechanism, which is a fusion of U-NetSED with Swin-GCE. The fusion process starts by concatenating the feature maps from the U-NetSED and Swin-GCE.

$$\mathcal{F}_{\text{combined}} = \text{Concat}(\mathcal{F}_{\text{U-NetSED}}, \mathcal{F}_{\text{Swin-GCE}}), \tag{3}$$

where $F_{\text{U-NetSED}}$ represents the local detailed feature map and $F_{\text{Swin-GCE}}$ is the globally contextualized features.

$$\mathcal{F}_{\text{fused}} = \text{Conv3d}(\mathcal{F}_{\text{combined}}). \tag{4}$$

Following concatenation, the combined features are processed through a fusion convolution layer. This convolutional layer performs the kernel operation to blend and compress the channel dimensions. The processed features undergo batch normalization and ReLU activation for a nonlinear feature transformation. To ensure the model's robustness and prevent overfitting, dropout is applied.

$$\mathcal{F}_{\text{activated}} = \text{ReLU}(BN(\mathcal{F}_{\text{fused}})). \tag{5}$$

To ensure the model's robustness and prevent overfitting, dropout is applied to $\mathcal{F}_{\text{activated}}$:

$$\mathcal{F}_{\text{regularized}} = \text{Dropout}(\mathcal{F}_{\text{activated}}). \tag{6}$$

$$\mathcal{F}_{\text{out}} = \text{Conv3d}(\text{Conv3d}(\mathcal{F}_{\text{regularized}})). \tag{7}$$

The layers of the network that come after the fusion further optimize the combined features, changing the dimensionality of the features while preserving spatial details. It results in a very detailed segmentation map, $\mathcal{F}_{\text{out}}$. The final segmentation output is refined through additional convolutional layers (kernels = $3 \times 3 \times 3$ and $1 \times 1 \times 1$) to map the processed features to a final output. The entire feature fusion and segmentation synthesis process can be formalized as a sequence of transformation functions applied sequentially to the input features:

$$\mathcal{F}_{\text{out}} = \mathcal{T}_{\text{N}}(\ldots(\mathcal{T}_2(\mathcal{T}_1(\mathcal{F}_{\text{in}})))). \tag{8}$$

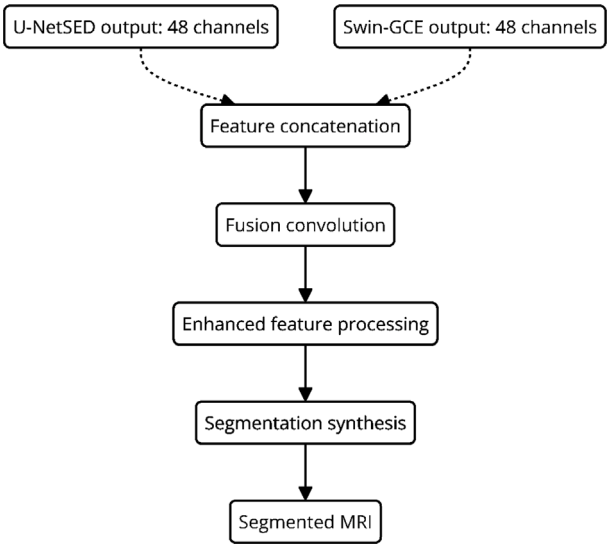

Fig. 5. A diagram illustrating the feature fusion and segmentation synthesis process in the FIGO framework. It shows the sequence of transformations applied to get the final segmentation output.

Table 3
Configuration details of the experimental environment.

Component	Specification
GPU	Nvidia RTX 6000 Ada
OS	Ubuntu 22.04
Memory	128 GB
CUDA version	Cuda 12.2
Python version	Python 3.11.5
Software	PyTorch 2.1.2
	MONAI 1.2.0
	NumPy 1.23.5

Individually, all the transform stages $\mathcal{T}_i$ are carefully produced. They all include convolution, batch normalization, activation, and dropout operations. The entire process of feature fusion, as well as the segmentation synthesis, reflects the architecture's innovative approach.

4. Implementation details and training protocol

4.1. Experiment setup

The model is executed in a Jupyter Notebook environment that allows for dynamic interaction at the development and testing levels. Transparency, reproducibility, and the ability to easily repeat or expand the work without deviation are the primary forces driving the experiment. The experiment is conducted using a workstation with Intel Xeon 4.8 GHz, 128 GB RAM, and Nvidia RTX 6000Ada GPU operated with Ubuntu 22.04, Cuda 12.2, Python 3.11.5, PyTorch 2.1.2, MONAI 1.2.0, and NumPy 1.23.5. For baseline methods, results were cited from their original works as shown in Table 4. For models with reproducible original code (e.g., UNet, TransUNet), as shown in Figs. 7 and 8, we implemented them under the same training setup as FIGO to ensure fairness and reproducibility. A learning rate of 1×10^{-4} was used with the Adam optimizer, dice loss, and a `ReduceLROnPlateau` scheduler (factor = 0.1, patience = 10). Training was performed using 3D data augmentations, including random rotations, flipping, and affine deformations. During inference, a sigmoid activation threshold of 0.6 was applied. All models were trained using mixed precision. This consistent setup allows for reliable performance comparison across methods without introducing bias in favor of the proposed architecture. Table 3 provides the hardware and software configuration details used in the experiment.

4.2. Inference process

The proposed model processes MRI data as full 3D volumes to capture the full spatial context of the data representation. This volumetric approach extends MRI analysis beyond the traditional slice-by-slice approach. By using 3D volumes, the model obtains high-resolution and detailed information about the lesion's volume, shape, and spatial correlation. This information is often not

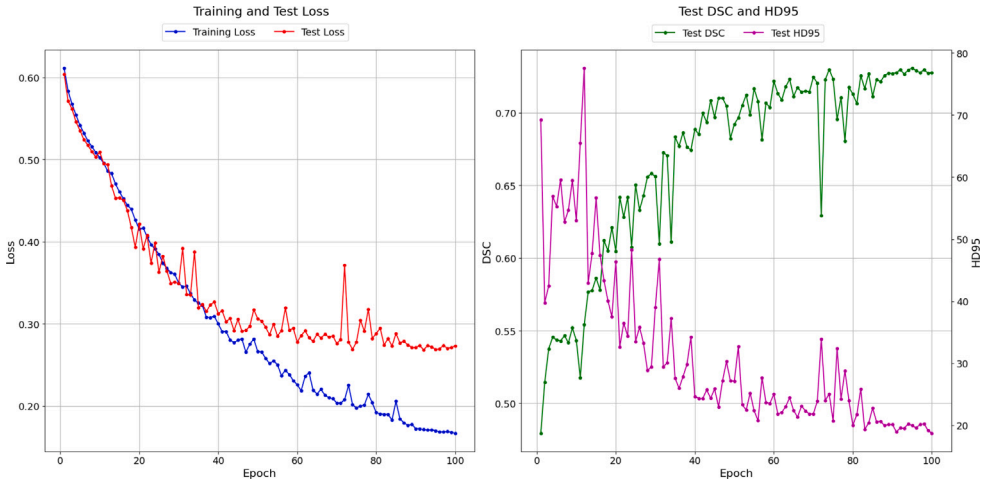

Fig. 6. The model performance across 100 epochs. The training loss that is gradually reduced shows the model is learning, while the test loss trend is also reducing, indicating that the model has generalization capability. The DSC demonstrates increasing accuracy in segmenting the desired regions. The Test HD95 shows a downward trend, indicating better precision in boundary segmentation.

as clear in slice-based analyses. Volumetric analysis is a key component that plays a critical role in clinical decision-making, such as determining the influence of lesions on overall brain function. By using 3D images, the model avoids the common challenges associated with slice-based methods, which tend to segment individual slices and may introduce segmentation errors due to the slice-by-slice treatment.

5. Experiments and results

The proposed model is a sequential integration of the spatial precision of U-Net and the contextual depth of SwinUNETR. The evolution of the dice score along with test loss and Hausdorff distance provides objective information on the model's ability to handle complex segmentation tasks (Fig. 6). A thorough assessment included comparing the model's results with other algorithms. When evaluated on the ATLAS v2.0 dataset, FIGO yielded optimal results: a test dice score of 0.731 with an average training time of 698.39 s per epoch. The model showed a stable increase in the dice score to precisely segregate stroke lesions, with an overall gain of 52.49% from its initial iteration. An improvement in the dice score shows a positive relationship with correct lesion identification that can promote effective patient care.

The HD95 fluctuated but demonstrated a decreasing tendency, which amounts to 72.99%, and an optimal value of 18.70. This metric is of the utmost importance for applications that need precise boundary detection and shows that the model has enhanced precision in the contouring of lesions. The ASSD also improved, decreasing by 96.30% to a final value of 1.269. It indicates the model's enhanced precision in boundary segmentation. The comparative analysis of the model with state-of-the-art (SOTA) is crucial for the positioning of FIGO. Table 4 shows the dice scores of each model categorized by their score nature, which shows the proposed model's performance with these methodologies. Compared to the previous models, the performance of FIGO and the adopted 3D TransUNet is highlighted.

FIGO achieves better segmentation accuracy by sequentially integrating deep learning structures without using any pre-trained models or post-processing. Table 4 shows that even though FIGO, despite being a 3D model, achieves better results in 3D, it also has a better dice score than the highest-performing 2D model. The analysis covers various techniques, including 2D models like SwinUnet and PSPNet, that often operate faster. However, they lack the depth perception required for lesion segmentation. The adapted 3D TransUNet achieves a dice score of 0.658 using volumetric data effectively. Furthermore, the 3D TransUNet has the lowest HD95 value, 32.96, and an ASSD value of 0.193. UNETR and 3D-ResU-Net also use volumetric data to provide the overall anatomical structures necessary for precise segmentation.

Models that use large-kernel attention approaches, such as LKA-E and LKA-ED, can balance computational efficiency and the ability to produce accurate segmentation results. One of the 3D residual U-Net variants that uses instance-wise and center-of-instance segmentation loss functions aims to improve the model's performance, considering the size disparities between the ischemic stroke lesions. The FISRG model, based on the fuzzy information seeded region growing method, has a high dice score of 0.942. Nevertheless, this is achieved with the use of single patient data. These factors, including detailed post-processing, even for one patient's data, led to the fact that FISRG is not included in the comparative study. This contributes to the growing understanding that such models need to be able to generalize across different clinical scenarios.

While Table 4 summarizes DSC values for various methods, most prior works lack boundary-aware metrics like HD95 and ASSD. To provide a more comprehensive benchmark, we reimplemented U-Net and 3D TransUNet. The reproduced UNet achieved DSC: 0.634, HD95: 24.59, ASSD: 17.72, while 3D TransUNet reached DSC: 0.658, HD95: 38.70, and ASSD: 40.74. Compared to

Table 4
Comparative analysis of SOTA on ATLAS v2.0.

Model	DSC ↑	Scoring	Performance analysis
X-Net [47]	0.313	2D	Moderate complexity, prone to overfitting
UNETR [47]	0.347	3D	High parameters, lower efficiency in 3D tasks
SwinUnet [47]	0.448	2D	Efficient feature integration, complexity in training
Residual U-Net [27]	0.504	3D	Requires balancing between depth and computational load
3D-ResU-Net [47]	0.512	3D	Computationally intensive for 3D data
SegNet [47]	0.533	2D	Balances efficiency and performance, less detailed
PSPNet [47]	0.580	2D	Requires significant computational resources
Residual U-Net [32]	0.581	3D	High computation due to complex instance-wise loss functions
U-net Transformer [27]	0.583	2D	Transformer integration increases complexity
HarDNet [47]	0.591	2D	Optimized for speed, lack feature extraction
U-Net [47]	0.598	2D	Limited by simplicity in handling complex patterns
Ensemble (PP) [33]	0.667	3D[a]	Complex ensemble, intensive post-processing
LKA-ED [28]	0.678	3D[a]	Optimization needed for varying kernel sizes
LKA-E [28]	0.682	3D[a]	Balances efficiency with performance
HCSNet [25]	0.697	3D[a]	Specialized for small lesion detection, high complexity
SQMLP-net [26]	0.709	3D[a]	Multi-tasking increases model complexity and training time
FIGO	**0.731**	3D	*This study*

↑ Indicates that higher is better.
[a] Average DSC.

these baselines, our FIGO model outperformed them across all metrics (DSC: 0.731, HD95: 18.70, ASSD: 03.87), demonstrating the effectiveness of our integrated spatial-contextual architecture and segmentation synthesis module. Figs. 7 and 8 display the visual results of various models used for stroke segmentation. They show an overview and a detailed close-up look, respectively.

The precision of the FIGO framework in segmenting small, large, and multiple lesions is good, as it has shown a close match with the annotations of experts. Such correspondence suggests that the model can detect clinically different changes in the neuropathological state. The model has demonstrated the ability to precisely locate small lesions, which are more difficult to detect than larger lesions due to their low contrast and signal-to-noise ratio. The model's specificity stands out in lesion-free subjects, a particularly essential factor used in the prevention of false positives and in avoiding the risk of wrong diagnosis. Furthermore, the model can correctly identify multiple lesions that might be present in a single case, which is necessary for a complete evaluation of the stroke lesions.

The visualization results also reveal the effectiveness of integrating SwinUNETR into the proposed model. SwinUNETR, which provides global context, significantly improves the segmentation of lesions. In cases with multiple lesions, the proposed framework also yields better accuracy in segmenting the lesions. This improved performance highlights the importance of incorporating SwinUNETR into more advanced fusion and segmentation synergistic approaches for better lesion identification in complex medical images. In the quantitative metrics, U-Net shows the optimal dice score of 0.633, a 13.29% decrease compared to the proposed model. The HD95 also decreased from 18.70 to 23.74, demonstrating the efficacy of incorporating SwinUNETR to improve model performance.

The proposed model tends to have technical limitations when delineating lesions with non-distinct edges or non-standard diffusion patterns, where the segmentation may unconsciously underestimate the actual size of the lesion. In these situations, the segmentation scenarios may deviate from the reality of the task of distinguishing pathological tissue from the brain's natural textural background. Such deviations, on the other hand, point out that the algorithmic interpretation of the same brain lesions is closely tied to their complex morphological profiles, indicating room for model improvement. The emergence of the above complexity implies a demand for larger data sets to train with more paths of pathological variants. The inclusion of more data with multiple modalities and potentially unusual appearances can improve the model's learning process. Sophisticated training would help FIGO become acquainted with the unpredictable variability in medical imaging, allowing it to generalize more effectively. This will bring about more precise segmentation outputs that can precisely depict the complexities of cerebral abnormalities.

5.1. Impact of pre-processing pipeline

This section analyzes the effect of two pre-processing pipelines on the segmentation of stroke lesions using a FIGO framework. The comparison evaluates the model's performance using a comprehensive and basic pre-processing approach. This study utilized fifty-two subjects to ensure a thorough assessment, which comprised a wide variation of stroke lesion characteristics. The smaller dataset was chosen to guarantee a representation of different lesion types and dimensions, accompanied by various pathologic homogeneities.

The comprehensive pre-processing included a well-defined protocol that comprised resampling, bias field correction, skull stripping, resizing, intensity normalization, and data augmentation and standardization. The model yielded a dice score of 0.74 with the comprehensive pipeline. However, this study revealed that the skull stripping process introduced noteworthy mistakes in accurately determining the cerebral limits on the 3D MRI. The distinguished detail in the brain shape in the 3D brain mapping algorithms made it difficult to differentiate the brain tissue from other non-brain structures accurately. Uniformly applying a set of

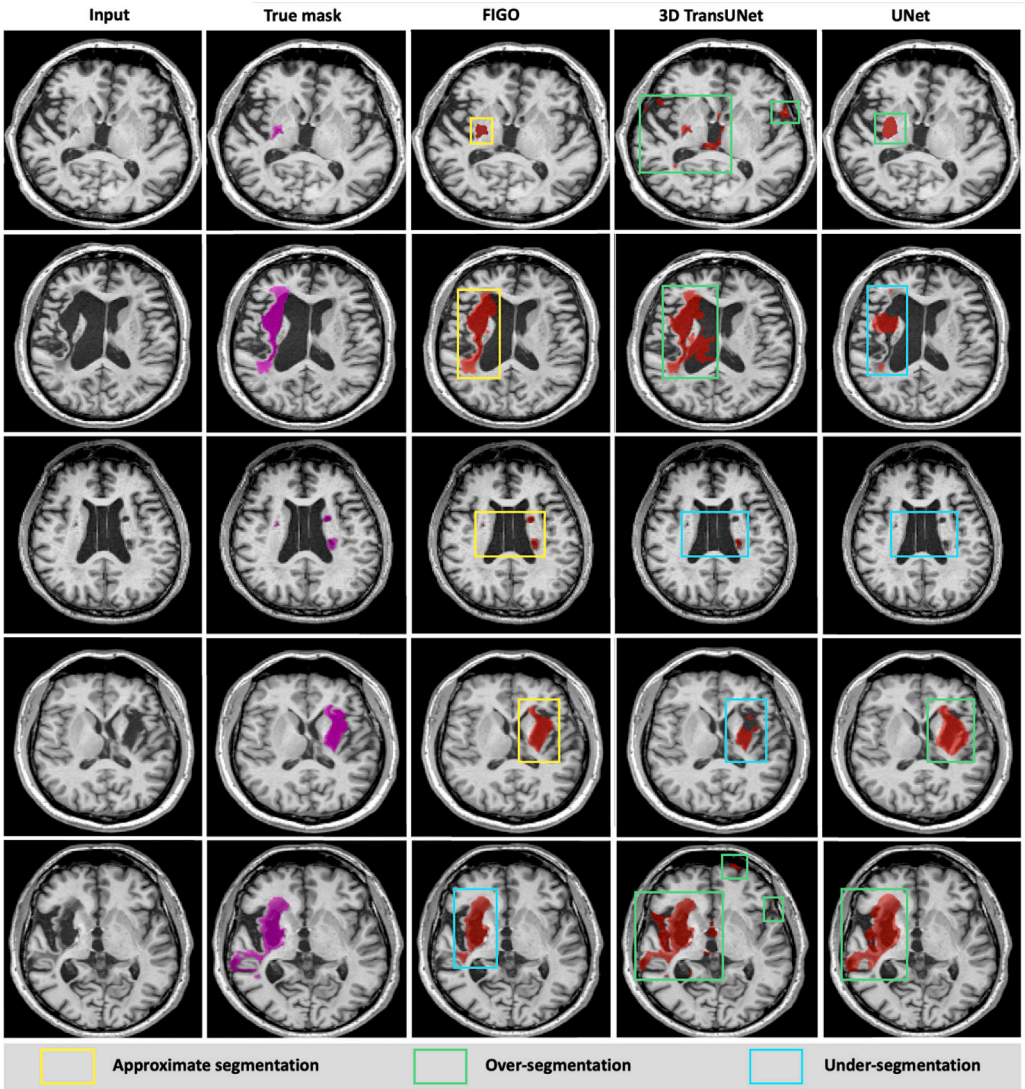

Fig. 7. The visualization depicts a systematic evaluation of FIGO, 3D TransUNet, and UNet performance on 3D MRI. Each row starts with the MRI, and then displays a true segmentation mask and the predictions made by each model. Yellow, green, and cyan boxes indicate approximate over-segmentations and under-segmentations, respectively.

all parameters for skull stripping across all subjects could lead to the unintended removal of important brain structures. To ensure accuracy, each subject must be processed individually for skull stripping, which is a time-consuming and challenging process. Thus, it is deemed prudent to exclude skull stripping from the comprehensive pre-processing scheme for the full training dataset, despite its inclusion in this subset. The basic pre-processing involves only the essential processes, with resampling, bias adjustment, and skull stripping purposefully left out. The model achieved the best dice score of 0.65 using the basic pre-processed data, indicating that the model's accuracy was reduced without comprehensive pre-processing.

5.2. Ablation studies

To analyze FIGO's architectural components, we conducted ablation experiments under identical training and evaluation conditions. The study isolated and evaluated the roles of: (i) spatial encoding via UNet-SED, (ii) global anatomical context modeling via Swin-GCE, and (iii) the feature fusion and segmentation synthesis module that integrates both streams. Table 5 summarizes the quantitative performance.

Each module in FIGO is functionally essential and contributes uniquely to the model's segmentation capacity. UNet-SED captures low-level spatial details through hierarchical convolutional operations but lacks contextual awareness. It leads to under-segmentation in spatially diffuse lesions and disconnected regions. This results in a low Dice and high ASSD, indicating imprecise boundary

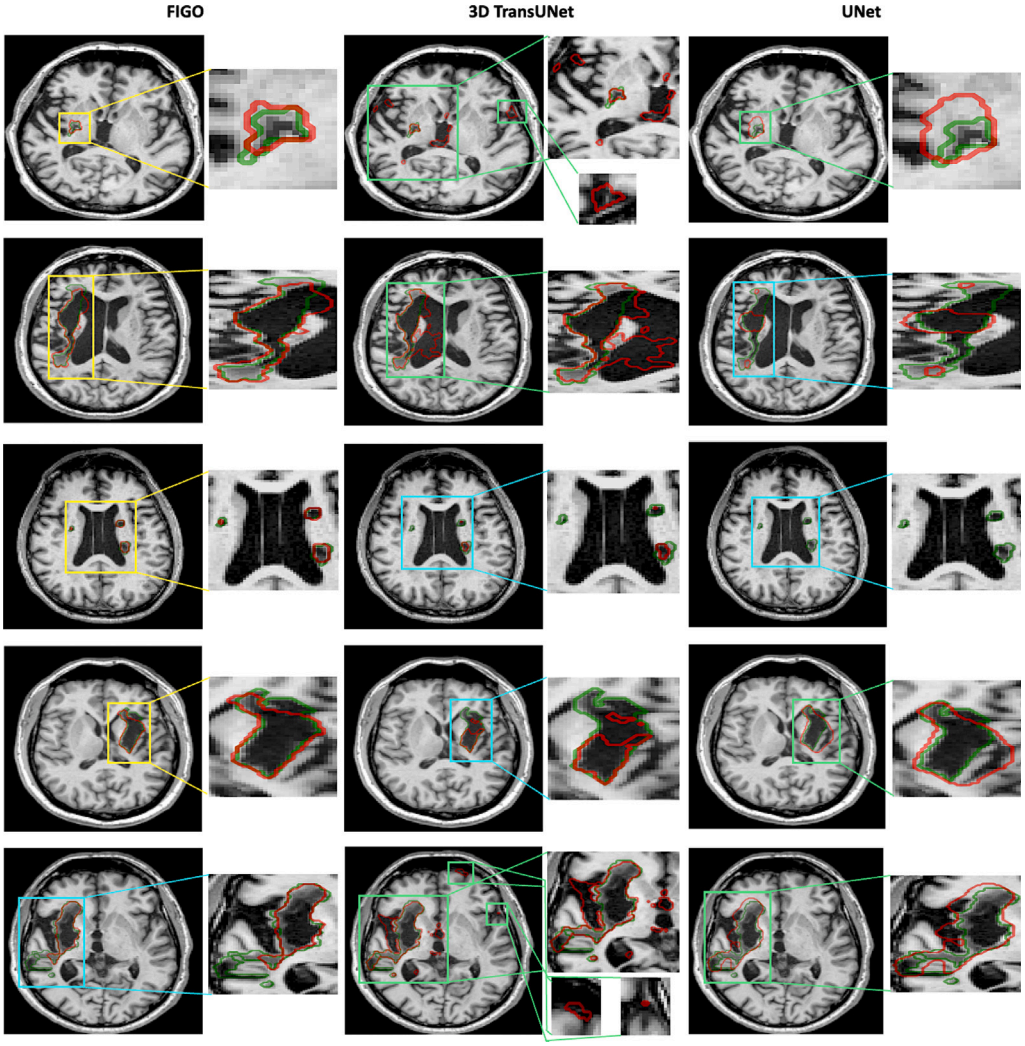

Fig. 8. The figure presents a detailed visualization that highlights key segmentation differences between models. Each zoom-in highlights regions where segmentation accuracy significantly differed. It is emphasizing the nuances between the true mask and model predictions.

Table 5
Ablation study of the FIGO.

Methods	DSC ↑	HD95 ↓ (mm)	ASSD ↓ (mm)
UNet-SED only	0.648	29.76	31.15
Swin-GCE only	0.686	20.56	05.98
w/o fusion and synthesis module	0.675	26.68	08.90
FIGO (ours)	**0.731**	**18.70**	**03.87**

modeling and fragmented region predictions. Swin-GCE, utilizing transformer-based self-attention, outperforms UNet-SED in HD95 and Dice scores. However, it lacks localized spatial priors, causing misalignment near fine boundaries and moderate surface error. This suggests that transformer-based context alone is insufficient for high-fidelity anatomical segmentation.

Without the fusion and synthesis module, UNet-SED and Swin-GCE components remain, but the learned fusion block and dedicated decoder were removed. Features are combined through element-wise addition, weakening disentanglement and suppressing synergistic representation learning, as shown by degraded dice, HD95, and ASSD. This confirms fusion must be learned and anatomically informed to integrate heterogeneous features effectively. The FIGO model incorporates all three components and achieves the highest performance across all metrics. Its feature fusion and segmentation synthesis module integrates contextual and local features, preserving global shape and boundary integrity.

5.3. Model parameters

The model design is expanded by a detailed analysis of the parameters, which shows the vast learning capacity that contributes to its high segmentation accuracy. The network architecture, with 100,076,263 trainable parameters, is a testament to the detail and complexity required to achieve results. The distribution of model parameters is also considered, with careful layer allocation allowing for highly accurate feature extraction and representation. The initial layers, such as unet.model.0.conv.unit0.conv, containing parameters in the hundreds, setting the stage for the model's feature learning. As the depth increases, so does the parameter count, culminating in layers with millions of parameters. FIGO, with 100 million trainable parameters, performs 2731.97 GMACs per forward pass. It achieves an average inference time of 5.22 s per 3D volume (1.9 volumes per second) and peak GPU memory usage of 8.71 GB on a single NVIDIA RTX 6000 Ada GPU. Compared to UNet and 3D TransUNet, FIGO's dual-path encoders and fusion module introduce modest complexity but remain computationally tractable and memory-efficient, supporting clinical-scale deployment.

6. Discussion and conclusion

The purpose of this research is to conduct a detailed analysis of the proposed framework in terms of its ability to segment stroke lesions from 3D MR images with high precision. The research focused on assessing the architecture and pre-processing strategies used to improve the model's segmentation accuracy. The model improves integration with a novel feature fusion strategy that balances local and global features. This approach emphasizes the significance of specialized feature fusion in managing the intricate details of medical imaging. It handles complex lesion morphologies more effectively than traditional methods, which lack a specialized mechanism for optimizing feature fusion and focus on 2D image processing.

The outcome of this study shows that FIGO achieved a test dice score of 0.731 using a volumetric approach that bypasses traditional post-processing, 2D, and 3D models. The proposed model performs well in detecting small, medium, large, and multiple lesions in a subject by sequentially integrating U-Net and SwinUNETR using advanced feature fusion and segmentation synthesis techniques. The spatial features of the U-Net improved the global contextual ability of SwinUNETR to detect large and multiple lesions in a subject. This approach reflects the natural anatomical complexity of the brain, making the model accurate and viable for segmenting stroke lesion over a wide range of appearances. This execution displays the FIGO's effectiveness in comparison to recent techniques used in similar tasks. The main idea The findings provide insight into various data processing pipelines and advanced neural network designs for accuracy in clinical diagnostics. These not only extend the current understanding of neural network applications in medical imaging but also underline the importance of these factors. The research found specific situations where the model's segmentation precision is still lacking, showing that more data can be added to the training dataset to boost the model's ability to analyze complex medical images. The ATLAS v2.0 dataset is a clinically comprehensive and diverse benchmark for ischemic stroke lesion segmentation. But validation for the FIGO paradigm on additional datasets is crucial for wider assessment of generalizability. Diversity in ATLAS v2.0 permits rigorous testing under real-world variability while keeping consistency in annotation. Deliberately selected for methodological isolation and reproducibility, ATLAS v2.0 is the focus of our ongoing research. We intend to expand the evaluation for the FIGO to publicly available benchmarks like ISLES and in-house datasets under different imaging protocols, voxel resolutions, and different lesion types. Such initiatives are in addition to the existing study based on benchmarks and provide for systematic testing of cross-domain robustness. They are also by our goal of improvement in generalizability through meta-learning, few-shot adaptation, and multimodal extensions in future iterations for the FIGO paradigm.

Furthermore, improving interpretable methodologies to understand the model's decision-making process could reveal the driving mechanisms of segmentation performance. Based on the data, the design imperfections can be identified and used for the next generation of architecture and to assist the clinician in interpreting the results.

CRediT authorship contribution statement

Muhammad Nouman: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Mohamed Mabrok:** Methodology, Data curation, Writing – review & editing, Funding acquisition. **Mohammad al-Shatouri:** Validation, Data curation, Project administration. **Essam A. Rashed:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The ATLAS v2.0 dataset used in this study is publicly available at https://github.com/npnl/atlas (visited 1 December 2024). The complete implementation of the FIGO framework, including model architecture, training pipeline, evaluation scripts, and visualization tools, are available from the first author on reasonable request.

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